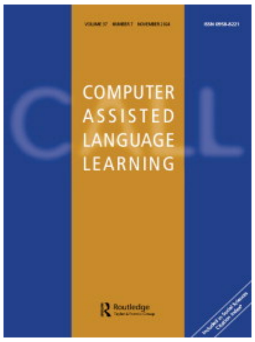

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A meta-analysis of the relative effectiveness of technology-enhanced language learning on ESL/EFL writing performance: retrospect and prospect

Masumeh Sadat Seyyedrezaei[a], Mohammad Amiryousefi[a], Ana Gimeno-Sanz[b] and Mansoor Tavakoli[a]

[a]Department of English, Faculty of Foreign Languages, University of Isfahan, Isfahan, Iran; [b]Department of Applied Linguistics, Universitat Politècnica de València (UPV), Spain

ABSTRACT

In recent years, there has been an upsurge of interest in the use of computer technologies for L2 writing instruction. Many researchers have empirically investigated the effects of the applications of educational technologies on language learners' writing performance and have highlighted the effectiveness of the use of educational technology in writing classes (Howell et al., 2021; Lan et al., 2019; Rahimi & Fathi, 2021). The aim of the present meta-analysis was threefold: (1) to examine the overall effectiveness of the applications of educational technology on English as a Second/Foreign Language (EFL/ESL) writing performance, (2) to investigate the substantive factors leading to between-study variation, and (3) to provide a genre-specific technology analysis. Sixty-four studies meeting the inclusion criteria were synthesized in this review. The results revealed that the applications of technology produce a large positive effect ($g = 1.00$) on EFL/ESL learners' writing performance. Moderator analyses, which were conducted with some study features, also led to the identification of two statistically significant moderator variables, i.e., genre of writing and type of technology. The results of meta-regression showed that there is a significant relationship between type of technology, genre of writing, and the overall effect size of the applications of educational technology. Pedagogical implications are discussed, and the future prospects for the use of technology-enhanced language learning (TELL) in the area of ESL/EFL writing instruction are presented.

Abbreviations: Bb: Blackboard; CLIL: Content and Language Integrated Learning; EAP: English for Academic Purposes; EFL: English as a Foreign Language; ESL: English as a Second Language; ESP: English for Specific Purposes; SCMC: Synchronous computer-mediated communication; TBLT: Task Based Language Teaching; TELL: Technology-Enhanced Language Learning; TESOL: Teaching English to Speakers of Other Languages; ZPD: Zone of Proximal Development

KEYWORDS
Technology-enhanced language learning; writing performance; meta-analysis

CONTACT Dr. Amiryousefi M.Amiryousefi@fgn.ui.ac.ir Department of English, Faculty of Foreign Languages, University of Isfahan, Isfahan, Iran.

1. Introduction

Recently, a growing interest has emerged in the use of innovative technologies to facilitate L2 writing instruction (Howell et al., 2021; Williams & Beam, 2019; Xu et al., 2019). Research has shown that the applications of educational technology can decrease L2 learners' writing apprehension (Bailey & Cassidy, 2019; Marandi & Seyyedrezaie, 2017); increase their attentional capacity; and, hence, provide learners with a good opportunity to focus on high-level processes such as 'organization of ideas, meta-cognitive evaluation, and peer feedback' (AbuSeileek & Abualsha'r, 2014, as cited in Chang et al., 2021, p. 2). Additionally, the use of technologies can reduce L2 learners' cognitive load and improve different aspects of L2 writing such as mechanical accuracy (spelling), compositional organization, and grammatical accuracy through the use of input, organizational, and evaluation tools, which educational technologies provide language learners with (Chang et al., 2021).

While there have been many studies on the applications of educational technology in writing instruction (Bikowski & Vithanage, 2016; Lan et al., 2019; Rahimi & Fathi, 2021), meta-analyses are needed to synthesize research evidence. To the authors' knowledge, only one meta-analytic study (Xu et al., 2019) has been conducted to systematically evaluate the effectiveness of the applications of educational technology on L2 writing quality as a subdomain. Xu et al.'s (2019) meta-analysis evaluated sixteen research articles published between 2000 and 2017 on the L2 writing performance of adult language learners. Their findings revealed that technology-enhanced writing instruction is more effective than non-technology instruction for adult language learners. They also posited that the effectiveness of the applications of educational technology can be mediated by program intensity and genre of writing.

While Xu et al. (2019) have deepened our understanding, several issues still remain open for further analysis, particularly the interaction effects between type of technology (collaborative/non-collaborative) and genre of writing. In fact, a meta-analysis is needed to reveal which type of technology is more effective in improving specific writing genres (e.g., argumentative, descriptive, narrative, etc.). Additionally, writing-specific studies are scarce, and the existing meta-analysis includes writing as a subdomain. Xu et al. (2019) also call for another meta-analysis to synthesize studies investigating the participants with different age ranges, i.e., children, teenagers (digital natives), and adults to find out whether the effectiveness of the applications of educational technology can be moderated by the age of learners. The reason why age range should be examined lies in the difference between digital natives and adults in terms of familiarity with technology; children and teenagers are usually familiar with and immersed in digital environments. Therefore, some researchers believe that the applications of educational technology can be more effective for digital natives than the older generations (Magsamen-Conrad & Dillon, 2020; Puebla et al., 2021). Conversely, some researchers (Kennedy et al., 2008; Lai & Hong, 2015) argue that the generational preference for technology does not always determine the success of the use of educational technology.

To bridge the gap, the present meta-analysis aimed at examining the overall effectiveness of the applications of educational technology on EFL/ESL writing performance by first investigating the moderating role of substantive factors such as age, genre of writing, level of L2 proficiency, and education level; and second analyzing the interaction effects between type of technology and genre of writing.

1.1. The advent of technology in writing classes

Writing is a language skill that is essential to academic success. Since it is an active and productive skill, students learning to write in a foreign language face multiple challenges (Alkhalaf, 2020). One of the challenges is that writing is a complex skill requiring a certain level of linguistic knowledge, i.e., vocabulary, grammar, and writing conventions (Ferris & Eckstein, 2020). Another challenging factor is that conventional writing classes are not motivating and suitable for language learners with different learning styles and abilities (Rybushkina & Krasnova, 2015). In the past few years, some researchers (Howell et al., 2021; Lin et al., 2020; Tananuraksakul, 2014; Williams & Beam, 2019) have emphasized that the applications of technology in writing instruction may tackle the challenges faced in conventional face-to-face writing classes, and have attributed the effectiveness of educational technologies to their less anxious-making nature, which reduces the learners' writing apprehension. Furthermore, the second generation of web-based technologies (Web 2.0) afford participation and collaboration at an unprecedented level and enable students to provide real-time feedback to each other; an opportunity that is often missing in traditional classroom settings due to the difficulty of providing real-time feedback and support to every student in a class period (Ferris & Eckstein, 2020).

Many studies have highlighted the potential applications of such innovative technologies in writing instruction (Bailey & Judd, 2018; Ebadi & Rahimi, 2018; Hsu & Lo, 2018; Seyyedrezaei et al, 2021; Tsai, 2015; Wang, 2015). For instance, Ebadi and Rahimi (2018) examined the effects of WebQuest-based instruction on EFL learners' critical thinking and academic essay writing. Their study revealed that the

WebQuest-based classroom was more effective and efficient than the face-to-face classroom in developing the learners' academic writing skills both in the short and long terms. They attributed the effectiveness of WebQuest to the fact that learners have access to an abundance of relevant materials in this environment, and to the fact that this environment can activate L2 learners' higher-level thinking skills in general and critical thinking skills in particular, which can subsequently contribute to the academic essay writing. Hsu and Lo (2018), conversely, concluded that technology-enhanced instruction significantly improved learners' argumentative essay writing and provided better results in terms of content and accuracy. Similarly, the findings of Wang's (2015) study revealed that the applications of technology not only develop learners' writing skills but also increase their motivation and interest in writing. Moreover, Tsai (2019) conducted a comparative study on the effectiveness of the application of a Blackboard (Bb) course management system versus non-technology instruction on L2 writing performance. The findings of her study revealed that Blackboard (Bb) course management system significantly influenced learners' writing through the provision of technical support and the user-friendly nature of this environment.

Although the researchers of the aforementioned studies have justified the effectiveness of the applications of educational technology in writing instruction, some research has reported no statistically significant difference between technology and non-technology instruction (Lin, 2014; Qi-Yuan, 2013; Topacio, 2018). For instance, Topacio (2018) mentioned that learners may not favor technology-mediated environments, especially online environments, due to the technical problems these environments may have, the distracting nature of these environments which makes concentration difficult, and the apprehension caused by students' lack of familiarity with specific kinds of technology. Lin (2014) also reported the results of a comparative study between blog-integrated instruction versus non-technology instruction in Taiwan with a particular focus on the effectiveness of blogs in improving learners' writing performance and motivation. His findings revealed that both instructional environments were equally effective. Lin (2014), however, asserted that language learners did not feel comfortable posting their writing assignments in blogs due to an open-access format of blogs; i.e., all of their classmates had access to their writings. He argued that this might prevent the learners from blogging freely and in turn, can lead to a group of less motivated writers. The contradictory findings of these studies motivated this meta-analysis to reach a general conclusion about the effectiveness of the applications of technology on writing performance.

1.2. Previous meta-analyses in the field of CALL

Several meta-analyses have reported the overall effectiveness of technology-enhanced language learning programs. Such meta-analytic studies have examined the effects of technology on the development of language literacy in general and different language skills and subskills in particular (e.g., Grgurović et al., 2013; Lee et al., 2020; Lin et al., 2013; Yun, 2011; Zhao, 2003). Lee et al. (2020) study can be considered one of the most recent meta-analyses, in which the authors calculated the effect sizes of 42 studies on K-12 literacy development in both conventional and technology-mediated environments. The findings of their study indicated a positive, medium effect size (+.47) in favor of technology-enhanced instruction. Zhao (2003), conversely, conducted a meta-analysis of nine studies focusing on the effectiveness of the applications of technology in language learning. The results of his study indicated a large effect size (+1.12), showing the positive effect of technology on language learning. Moreover, the findings of his study revealed that the use of technology generally yielded more effective outcomes when compared to conventional instruction. In the same way, Lin et al. (2013) analyzed ten articles and doctoral dissertations to examine the overall effectiveness of text-based synchronous computer-mediated communication (SCMC) on language learning. They reported a small but positive effect size (+.33), indicating that text-based SCMC can improve language learning more than other means of communication. Similarly, Grgurović et al. (2013) conducted another meta-analysis and analyzed 85 articles. They compared language learning with and without the applications of educational technology. The findings of their study indicated a small but positive effect in favor of technology-enhanced instruction in studies using rigorous research designs. They found that technology-enhanced instruction was at least as effective as non-technology instruction.

In addition to the aforementioned meta-analyses indicating the overall effectiveness of technology-enhanced language learning (TELL), other meta-analyses examined the effects of educational technologies on specific skills or subskills, i.e., reading comprehension, vocabulary learning (Abraham, 2008; Cheung & Slavin, 2011), and speaking (Lin, 2016). On vocabulary learning, many meta-analysis studies have been conducted (Abraham, 2008; Taylor, 2006; Yun, 2011). In his study, Taylor (2006) analyzed eighteen studies that compared the effectiveness of traditional versus computer-assisted native language glosses on L2 reading comprehension. The findings of his study revealed that electronic glosses (e-glosses) significantly improved learners' L2 reading comprehension. Similarly, Abraham (2008) analyzed eleven research studies to investigate the effects of computer-mediated glosses on L2 reading comprehension and vocabulary learning. The findings of his study favored computer-mediated glosses with a medium effect (.73) and a large effect (1.40) on learners' reading comprehension and vocabulary learning, respectively. In addition, Yun (2011) conducted a meta-analysis to investigate the comparative effects of two types of hypertext gloss, i.e., text-only (single) versus text + visual (multiple) hypertext glosses, on L2 vocabulary acquisition in online contexts. In his study, he reported a positive moderate effect size (+.46), indicating that multiple hypertext glossing was more effective than single hypertext glossing. Considering reading comprehension, Cheung and Slavin (2011) meta-analysis focused on the effects of educational technology on learners' reading comprehension. The results favored educational technology with a positive though small (+.16) effect size over non-technology instruction. Lin (2016), conversely, compared the efficacy of L2 oral instruction with and without educational technologies. A total of twenty-five studies from 2000 to 2012 were analyzed. Her meta-analysis reported an overall medium effect size for computer technologies. These meta-analyses examined TELL from various perspectives and repeatedly referred to some substantive and methodological features as potential moderators that might differentiate the effectiveness of educational technologies. For instance, in her study, Lin (2016) revealed that methodological factors such as task type, assessment task, outcome measurement, and treatment length were significant moderator variables affecting the effectiveness of educational technologies.

On writing skills, a meta-analysis was conducted by (Little et al. 2018). They systematically evaluated the effectiveness of the applications of educational technology for L1 writing performance by synthesizing six academic articles published up to 2014. They exclusively focused on L1 writing. Their study reported an overall small effect size of .28, suggesting that technology has a small effect size for L1 writing quality. Subsequently, another meta-analysis was done by Xu et al. (2019), who reviewed sixteen empirical studies. They found that computer-mediated instruction and feedback had a large positive effect ($g = 1.28$) on adult learners' writing quality. While this synthesis on adult learners is valuable, a meta-analysis focusing on broader age ranges of learners will be beneficial for language practitioners and researchers to help them make proper decisions about the use of appropriate educational technology for each age group.

Although there are a few meta-analyses on the effectiveness of computer technology on general writing outcomes, up to the present, no meta-analysis has specifically considered the effect of TELL on ESL/EFL writing performance. As writing is an important skill in academic success, and online instruction has become obligatory during the corona pandemic (Khan & Hameed, 2021; Lim & Richardson, 2021), conducting a meta-analysis to understand how to deliver effective writing instruction is of paramount importance. Since Xu et al. (2019) meta-analysis reviewed the academic articles published until 2017, the empirical studies conducted during the corona pandemic were not included. Due to the rapid advancement of educational technologies and the necessity to use online instruction, a meta-analysis that reviews recently published articles is of great value.

1.3. Research questions

The following research questions guide the present meta-analysis:

1. What is the overall effectiveness of TELL on ESL/EFL writing performance?
2. To what extent are the effect sizes of the studies investigating the effect of TELL on ESL/EFL writing performance heterogeneous?
3. To what extent does the effectiveness vary across substantive and methodological features, i.e., research design, the genre of writing, proficiency level, type of technology, measurement type, treatment duration, sample size, and education level?
4. What are the combined moderating effects of genre of writing and the type of technology on the effectiveness of technology-enhanced writing instruction?

2. Method

2.1. Identification and retrieval of studies

The following electronic databases were examined to identify the studies on the relative effectiveness of TELL on ESL/EFL writing performance from 1990 to 2020:

1. The electronic databases and online search engines: Education Resources Information Center (ERIC), ProQuest, Linguistics and Language Behavior Abstract (LLBA), Springer Link, Web of Knowledge, Web of Science, and Google Scholar; and
2. Online research journals: CALICO, CALL, ReCALL, CALL-EJ, System, Educational Technology Research and Development, JSTOR, Language Learning and Technology, Journal of Educational Technology and Society, Ingenta, Foreign Language Annals, Innovation in language learning and teaching, TESOL Quarterly, Teaching English with Technology, TOJDE, Reading in a Foreign Language, SAGE Journals, The Modern Language Journal, Asian TEFL.

A combination of the following keyword concepts was used for the preliminary search: distance education, blended learning, online collaborative writing, online writing performance, writing and educational technology, online writing instruction, online writing environments, web-based writing, e-portfolio and writing, weblog, Google Docs, Etherpad, Edmodo, and so forth. Furthermore, the references of all the selected studies were hand-searched to retrieve articles that might have been lost in the search process. The search process resulted in identifying 384 studies in the domain of technology-enhanced writing instruction. Finally, all the retrieved studies were examined to include those meeting the inclusion criteria.

2.2. Inclusion/exclusion criteria

To be included in this meta-analysis, the following conditions were met for each study: (a) technology-enhanced EFL/ESL writing instruction was compared with non-technology EFL/ESL writing instruction; (b) studies were published between 1990 and 2020 and were written in English; (c) studies with quasi-experimental or experimental designs were included; (d) enough information for calculating effect sizes (mean and standard deviation of both groups, also difference in means, *p*-value, *t* or *f*, etc.) was provided; (e) studies measuring learners' writing performance quantitatively were included; and (f) studies contained a comparison group that did not receive technology-enhanced writing instruction were included.

Conference papers and the studies that used different outcome measurement procedures for experimental and control groups, used teachers' descriptive reports and self-assessment to examine writing performance, and investigated native speakers' writing performance were excluded.

2.3. Coding study features

After the screening stage, sixty-four studies (Appendix A) meeting the inclusion criteria remained to be coded for a variety of substantive features. To code study features, a detailed coding sheet was developed. According to Alsadhan (2011), a coding sheet should consider participants' characteristics, research design, and quantitative data to calculate the effect size. Thus, a detailed coding sheet (Appendix B) was developed based on the previous meta-analyses, including the following parts.

A. The context under which the studies were conducted:
   a. **Learning context.** Learning context was coded as 'EFL'/'ESL';
   b. **Types of educational technologies**. Types of educational technologies were coded as 'collaborative'/'non-collaborative'. According to Alharbi (2015), collaborative technologies are environments or tools that provide the learners with the opportunity to collectively create and edit the content and organization of their writing.

These environments increase collaborative engagement (Wichadee, 2013) and provide the opportunity to receive peers' feedback and comments (Zou et al., 2016);

c. **Genres of writing.** Genres of writing in this met-analysis were coded as 'descriptive', 'academic', 'argumentative', 'expository', 'narrative', 'summary', and 'mixed' writing. These codes were used by the authors of the included studies to describe the genres of writing involved. Although these genre types are academic, the coding sheet also included another category entitled 'academic genre'. This is because this category was used in a few included studies to refer to the genre type used. The studies involving different genre types were coded as 'mixed'.

B. Design of study:

a. **Sample size**. The sample size of more than sixty participants was coded 'large'. Conversely, the sample size of fewer than sixty participants was coded 'small';

b. **Research design**. The research design was coded as 'true-experimental' and 'quasi-experimental';

c. **Learner variables**. This category consisted of two parts, i.e., proficiency level and education level. Learners' proficiency levels were coded as 'elementary', 'lower intermediate', 'intermediate', 'upper-intermediate', and 'advanced', and learners' education levels were coded as 'K-12', 'under-graduate', and 'graduate'.

d. **Treatment duration**. This category refers to the amount of time under which language learners received technology-enhanced writing instruction. The studies having a treatment duration of three months and more were coded as 'long' and less than three months as 'short'. The researchers considered a treatment duration of three months as the basis for their coding since this duration represents half of an academic year.

e. **Outcome measurement procedures**. This category was coded as 'researcher-made' and 'standardized test'.

To ensure inter-coder reliability, the included studies were independently examined and coded by two independent coders based on the coding sheet. The inter-coder agreement was 96%, which was an appropriate index for the purpose of the study.

2.4. Data analysis

2.4.1. Effect size calculation

In behavioral sciences, Cohen's d, Glass's Δ, and Hedges' g are used to estimate the effect size (Turner & Bernard, 2006). However, Cohen's d is preferred to Glass's Δ. 'Glass's Δ uses standard deviation of control group' (Turner & Bernard, 2006, p. 45). Whereas, in Cohen's *d* formula, the standard deviation of both control and experimental groups are considered. Conversely, Cohen's *d* is the best estimate in the meta-analysis of studies with large sample sizes, while 'Hedge's *g* is recommended for use in systematic reviews that include studies with both large and small samples' (Chalmers & Altman, 1995, as cited in Turner & Bernard, 2006, p. 45). Thus, in the present meta-analysis, Hedge's *g* was used to estimate the effect sizes for the included studies.

2.4.2. Publication bias

Publication bias also called 'funnel plot asymmetry' or 'file-drawer problem', is very common in the existing literature. It is so-called because 'positive results have a better chance of being published, are published earlier, and are published in journals with higher impact factors' (Dubben & Beck-Bornholdt, 2005, p. 433). Therefore, detecting and correcting publication bias are of high importance, especially in the meta-analysis, to reach reliable and generalizable conclusions.

To check the presence of publication bias in the present meta-analysis, the funnel plot was examined. In the funnel plot, the horizontal axis is the effect size plotted against the standard error located on the vertical axis. Generally, an asymmetrical distribution of effect sizes in the funnel plot indicates the presence of publication bias (Duval & Tweedie, 2000). However, it is difficult to interpret the funnel plot (Figure 1) in the present meta-analysis due to the existence of some studies outside the funnel. These studies are evidence of significant heterogeneity.

Thus, in addition to the funnel plot, Classic Fail-safe *N* and Orwin's Fail-safe *N* tests were used to assess the publication bias. Classic Fail-safe *N* and Orwin's Fail-safe *N* tests were run to determine how many studies with null results are required to nullify the overall effect size found in the present meta-analysis. The result of classic fail-safe *N* indicated that

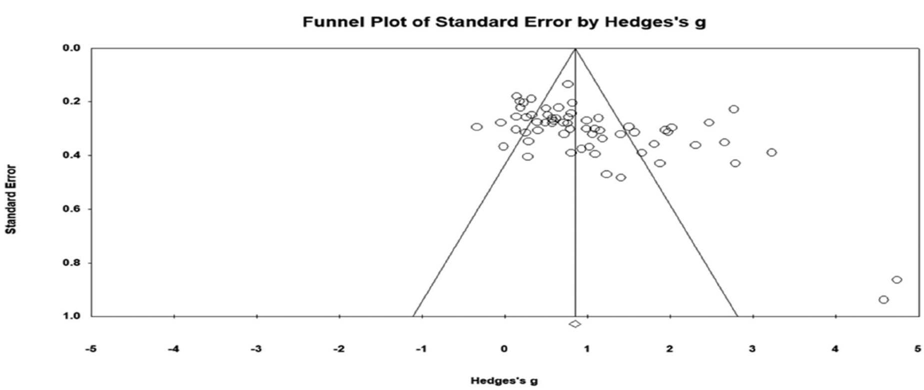


Figure 1. Publication bias.

1240 additional studies with non-significant results would be needed to nullify the overall effect size. Moreover, the result of Orwin's fail-safe *N* test revealed that 1019 null studies are required to make the overall mean effect size closer to a trivial level (g = .05).

3. Results

In this meta-analysis, a systematic literature search identified 132 studies that met the inclusion criteria. Subsequently, the identified studies were further scrutinized to exclude those which did not provide adequate and/or suitable data. As a consequence, the search resulted in 64 studies that met all the inclusion criteria and were considered in the present meta-analysis.

3.1. Overall effect size of TELL

To respond to the first research question estimating the overall effectiveness of TELL in EFL/ESL writing performance, the overall effect size was calculated using a random-effects model. According to Borenstein et al. (2009), a random-effects model should be assumed when the effect sizes of the included studies vary. Hence, in the present meta-analysis, a random-effects model was applied for two reasons. First, the analysis of Q statistics showed that there was substantial heterogeneity in the effect sizes, $Q(63)$ = 451.930, $p<.001$, I^2 = 86.06% (Table 1). Second, the included studies investigated the participants with different age ranges and proficiency levels.

As indicated in Table 1, the overall mean effect size of TELL under a random effect model was $g=1.00$, which is large according to Cohen's (1988) interpretation of the magnitude of effect size. So, it can be concluded that TELL had a large positive effect on EFL/ESL writing performance.

To assess whether the mean effect size is statistically significant, 95% confidence intervals were calculated for the weighted mean effect size. Generally, confidence intervals that contain the zero value show that there is no statistically significant difference between the effectiveness of experimental and control groups. Whereas 'the confidence intervals that do not include the zero value indicate that the observed effect differs probabilistically from the null hypothesis of no effect' (Norris & Ortega, 2000, p. 462).

Table 1. The overall effect size of TELL on writing performance.

	Weighted effect size			95% CI		Heterogeneity			
Model	*K*	*g*	*SE*	Lower	Upper	*Q*-value	*df*	*p*-value	I^2
Fixed	64	.846	.035	.778	.914	451.930	63	.000	86.060
Random	64	1.007	.095	.820	1.194				

As Table 1 demonstrates, the lower limit and upper limit equaled .82 and 1.19, respectively. Accordingly, the confidence interval range (.82 to 1.19) doesn't include zero. Thus, it can be concluded that TELL improved EFL/ESL writing performance, i.e., the learners receiving technology-integrated writing instruction outperformed those receiving traditional instruction.

The forest plot presented in Figure 2 indicates the effect size of each study.

As shown in Figure 2, the effect size of each study is represented by small boxes, and the confidence interval for each study is indicated by a horizontal line crossing each box. Moreover, on the right side of Figure 2, the weight of each study to the overall mean effect size is indicated by a square.

3.2. Sensitivity analysis and outlier analysis

To investigate whether there is any outlier leading to an extraordinary change in the overall mean effect size, one study-removal, proposed by Borenstein et al. (2009), was conducted. Since the overall mean effect size did not change ($g = 1.00$), it can be concluded that there is no outlier affecting the overall effect size (Table 2).

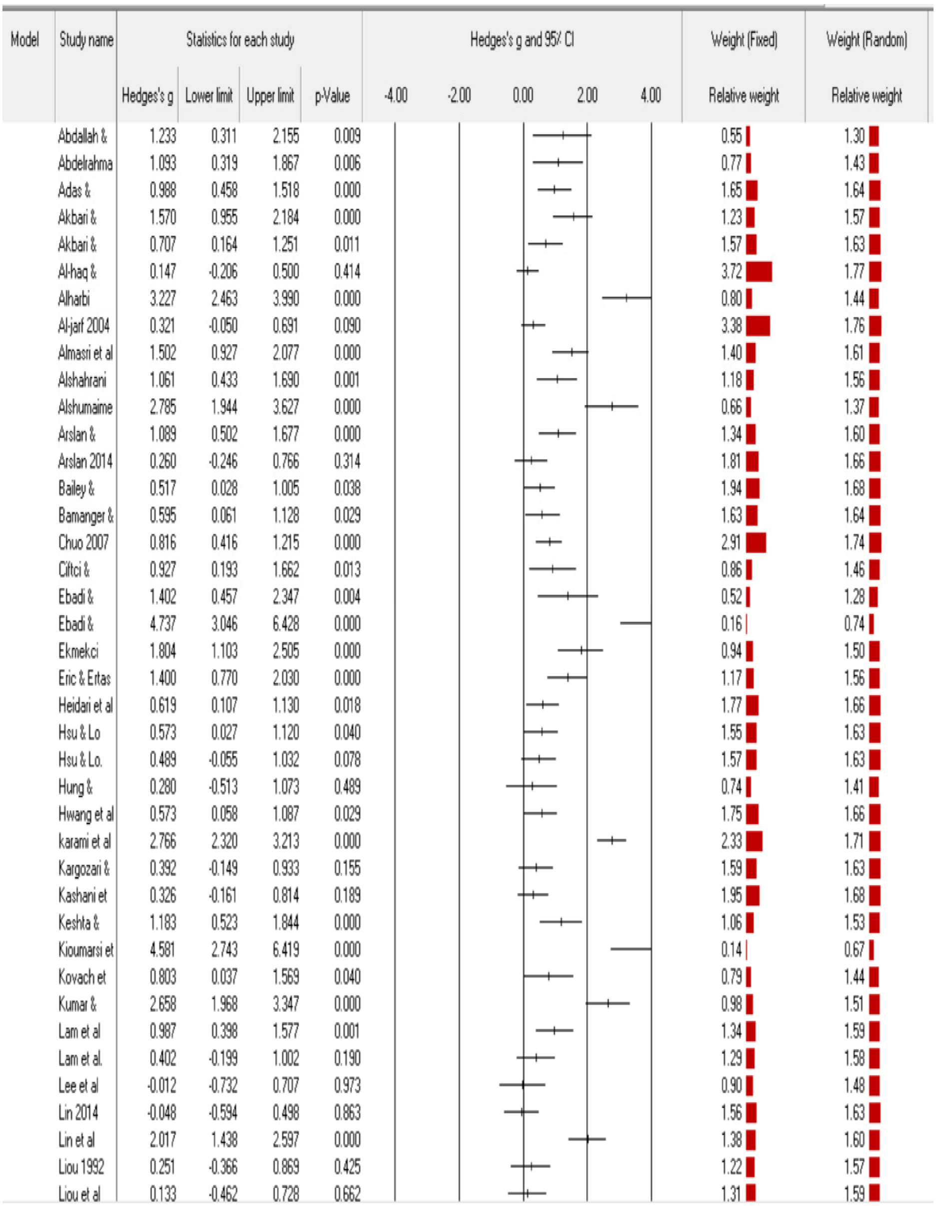
3.3. Heterogeneity of effect size

To answer the second research question, i.e., whether there is any significant heterogeneity among the effect sizes of the included studies, Cochran's test of homogeneity of variance (Q-test) was run. According to Borenstein et al. (2009), Q-test reveals whether all studies share a common effect size.

As indicated in Table 1, there is heterogeneity among the effect sizes of studies Q (63) = 451.930, $p < .05$. Moreover, the I^2 value of 64 effect sizes ($I^2 = 86.06$) revealed that approximately 86% of the observed variance is due to differences in true effect sizes, while only 14% of observed variance may be due to sampling error. According to Higgins and Thompson (2002) interpretation, the value of I^2 around 75% or larger shows a high level of heterogeneity and requires moderator analysis to clarify the variables leading to this variation. So, it can be concluded that there are some variables leading to the variation among the effect sizes of the studies.

3.4. Moderator analysis

In response to the third research question investigating the potential sources of such variation, moderator analysis was conducted. The methodological and substantive features of the studies were considered as moderator variables, i.e., the genre of writing, research design, type of technology, proficiency level, measurement type, treatment duration, sample size, and education level. Table 3 shows a summary of the results of the moderator analysis.



Model	Study name	Hedges's g	Lower limit	Upper limit	p-Value	Weight (Fixed) Relative weight	Weight (Random) Relative weight
	Abdallah &	1.233	0.311	2.155	0.009	0.55	1.30
	Abdelrahma	1.093	0.319	1.867	0.006	0.77	1.43
	Adas &	0.988	0.458	1.518	0.000	1.65	1.64
	Akbari &	1.570	0.955	2.184	0.000	1.23	1.57
	Akbari &	0.707	0.164	1.251	0.011	1.57	1.63
	Al-haq &	0.147	-0.206	0.500	0.414	3.72	1.77
	Alharbi	3.227	2.463	3.990	0.000	0.80	1.44
	Al-jarf 2004	0.321	-0.050	0.691	0.090	3.38	1.76
	Almasri et al	1.502	0.927	2.077	0.000	1.40	1.61
	Alshahrani	1.061	0.433	1.690	0.001	1.18	1.56
	Alshumaime	2.785	1.944	3.627	0.000	0.66	1.37
	Arslan &	1.089	0.502	1.677	0.000	1.34	1.60
	Arslan 2014	0.260	-0.246	0.766	0.314	1.81	1.66
	Bailey &	0.517	0.028	1.005	0.038	1.94	1.68
	Bamanger &	0.595	0.061	1.128	0.029	1.63	1.64
	Chuo 2007	0.816	0.416	1.215	0.000	2.91	1.74
	Ciftci &	0.927	0.193	1.662	0.013	0.86	1.46
	Ebadi &	1.402	0.457	2.347	0.004	0.52	1.28
	Ebadi &	4.737	3.046	6.428	0.000	0.16	0.74
	Ekmekci	1.804	1.103	2.505	0.000	0.94	1.50
	Eric & Ertas	1.400	0.770	2.030	0.000	1.17	1.56
	Heidari et al	0.619	0.107	1.130	0.018	1.77	1.66
	Hsu & Lo	0.573	0.027	1.120	0.040	1.55	1.63
	Hsu & Lo.	0.489	-0.055	1.032	0.078	1.57	1.63
	Hung &	0.280	-0.513	1.073	0.489	0.74	1.41
	Hwang et al	0.573	0.058	1.087	0.029	1.75	1.66
	karami et al	2.766	2.320	3.213	0.000	2.33	1.71
	Kargozari &	0.392	-0.149	0.933	0.155	1.59	1.63
	Kashani et	0.326	-0.161	0.814	0.189	1.95	1.68
	Keshta &	1.183	0.523	1.844	0.000	1.06	1.53
	Kioumarsi et	4.581	2.743	6.419	0.000	0.14	0.67
	Kovach et	0.803	0.037	1.569	0.040	0.79	1.44
	Kumar &	2.658	1.968	3.347	0.000	0.98	1.51
	Lam et al	0.987	0.398	1.577	0.001	1.34	1.59
	Lam et al.	0.402	-0.199	1.002	0.190	1.29	1.58
	Lee et al	-0.012	-0.732	0.707	0.973	0.90	1.48
	Lin 2014	-0.048	-0.594	0.498	0.863	1.56	1.63
	Lin et al	2.017	1.438	2.597	0.000	1.38	1.60
	Liou 1992	0.251	-0.366	0.869	0.425	1.22	1.57
	Liou et al	0.133	-0.462	0.728	0.662	1.31	1.59

Figure 2. Forest plot and relative weights of the studies.

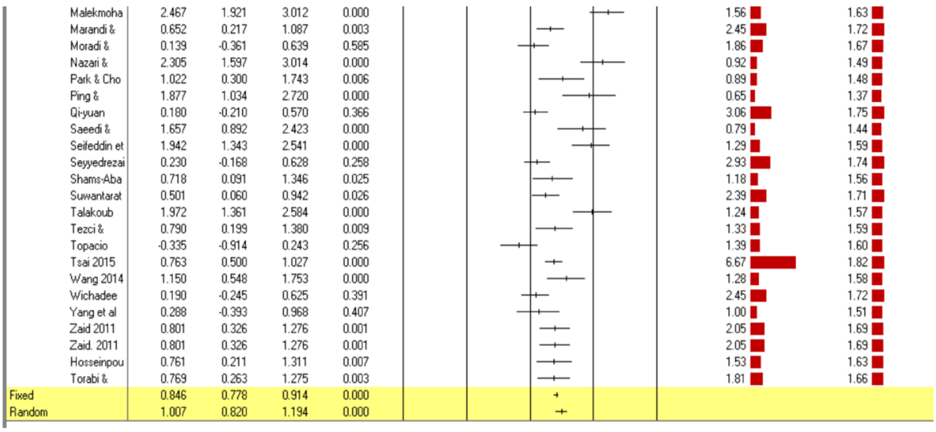


Figure 2. Continued.

Table 2. Outlier analysis.

	Weighted effect size			95% CI		Heterogeneity			
Model	*K*	*g*	*SE*	Lower	Upper	*Q*-value	*df*	*p*-value	I^2
Fixed	64	.846	.035	.778	.914	451.930	63	.000	86.060
Random	64	1.007	.095	.820	1.194				

3.4.1. Research design

Different research designs might be the potential sources of variation in effect sizes (Abrami & Bernard, 2006). Regarding the research design, the studies included in the present meta-analysis were classified into true experimental ($N = 15$) and quasi-experimental ($N = 49$). True experimental studies referred to those in which the participants were randomly assigned into experimental (TELL) and control (non-technology) groups. Whereas, the quasi-experimental studies were those in which there was no random assignment.

As indicated in Table 3, the mean effect sizes for quasi-experimental and true experimental studies were .90 and 1.39, respectively. However, there was no statistically significant difference between the two research designs (Q-value= 3.09, p = .079). The result revealed that TELL had a large positive effect on both quasi-experimental studies and true experimental ones.

3.4.2. Genre of writing

Regarding the genre of writing, the included studies examined the effect of TELL on learners' different genres of writing; i.e., academic ($N = 4$, ES = .98), argumentative ($N = 22$, ES = 1.37), descriptive ($N = 7$, ES = .73), expository ($N = 2$, ES = .53), narrative ($N = 3$, ES = 1.10), summary ($N = 1$, ES= .19), and mixed genres ($N = 25$, ES = .85). According to moderator analysis (Table 3), there was a statistically significant between-group variation ($Q = 17.80$, p = .00).

Table 3. Moderator analyses.

Moderator variable	Subgroup	Number of studies	Effect size (Hedge's *g*)	Lower limit	Upper limit	*Q*-value	*p*-value
Genre of writing	Academic	4	.988	.103	1.873	17.807	.007
	Argumentative	22	1.372	.959	1.784		
	Descriptive	7	.739	.145	1.334		
	Expository	2	.531	.145	.916		
	Mixed	25	.855	.632	1.077		
	Narrative	3	1.108	.307	1.908		
	Summary	1	.190	−.245	.625		
Research design	Quasi-experimental	49	.903	.707	1.099	3.090	.079
	True experimental	15	1.392	.883	1.901		
Type of technology	Collaborative	34	1.197	.942	1.453	4.373	.037
	Non-collaborative	30	.799	.527	1.071		
Proficiency level	Advanced	10	1.017	.411	1.623	3.811	.577
	Elementary	11	1.026	.609	1.443		
	Intermediate	29	.951	.709	1.194		
	Lower-intermediate	5	.759	..082	1.436		
	Upper-intermediate	5	1.821	.867	2.775		
	Not reported		.767	.004	1.530		
Measurement type	Researcher-made	47	.936	.738	1.133	1.190	.275
	Standardized	17	1.210	.759	1.660		
Treatment duration	Long	37	.915	.675	1.156	1.278	.258
	Short	27	1.138	.836	1.440		
Sample size	Large	29	1.042	.756	1.327	.157	.692
	Small	35	.966	.6724	1.208		
Education level	K-12	14	1.053	.667	1.439	3.083	.214
	Undergraduate	40	.906	.674	1.137		
	Graduate	10	1.475	.870	2.080		

3.4.3. Type of technology

The studies were classified into two groups regarding the type of technology used; collaborative and non-collaborative technologies. The studies using educational technologies to provide the learners with appropriate opportunities to interact or collaborate with their teachers or peers were coded as collaborative technologies. Whereas, the studies using educational technologies or technology-enhanced environments in which there was no interaction were classified as non-collaborative. Generally, the technologies which are not applicable for collaborative text construction are considered as a non-collaborative writing environment. Additionally, the studies using technologies that are collaborative in nature but were used non-collaboratively in the studies were classified as non-collaborative.

Thirty-four studies ($g = 1.19$) examined the effectiveness of Google Docs, Wiki, Webquest, and Edmodo as online collaborative environments, and thirty studies (g = .79) investigated the effect of computer games as non-collaborative tools. The result revealed that the type of technology had a statistically significant effect on between-studies variation ($Q = 4.37$, $p = .03$). The finding indicated that collaborative tools led to a larger effect size than non-collaborative ones.

3.4.4. Proficiency level

To investigate whether the learners' proficiency levels had any statistically significant impact on the overall effect size, the studies were coded regarding five levels of language proficiency, i.e., elementary, lower intermediate, intermediate, upper-intermediate, and advanced. Among the included studies, twenty-nine had focused on intermediate learners, eleven studies on elementary learners, ten studies on advanced learners, five studies on lower-intermediate learners, and five studies on upper-intermediate learners. The result revealed that TELL had large effects on upper-intermediate ($g = 1.82$), advanced ($g = 1.07$), intermediate ($g = .95$), and elementary ($g = 1.02$) learners, and moderate effects on lower-intermediate learners ($g = .75$). However, there was not any statistically significant difference between different levels of proficiency ($p = .57$).

3.4.5. Measurement type

In 47 studies, the participants' writing performance was measured by a researcher-made test. In the remaining seventeen studies, the writing outcomes were measured by a standardized test. The effect sizes for researcher-made and standardized tests were .93 and 1.21, respectively. However, as Table 3 indicates, there was no statistically significant difference between the effect sizes of these two types of measurement ($Q = 1.190$, $p = .27$).

3.4.6. Treatment duration

To investigate the effect of TELL in different treatment durations, the studies were categorized into two subcategories; long and short duration. The studies, in which the treatment lasted three months or more were considered to be long ($N = 37$). Whereas, the treatment lasting less than three months was considered to be a short duration ($N = 27$). Although the effect size for studies that were less than three months (ES = 1.13) was larger than those studies that lasted longer than three months (ES = .91), the between-group variation was not statistically significant (Q-value = 1.27, $p = .25$).

3.4.7. Sample size

According to Slavin and Smith (2009), sample size can be considered as a potential source of variation in systematic reviews. To examine whether the number of participants can lead to variation in effect sizes of the present meta-analysis, the small sample size was considered. In this systematic review, there were a total of 29 studies with 60 or more students (g = 1.04) and 35 studies with fewer than sixty students (g =.96). The results indicated that TELL had large positive effects on both small and large numbers of participants. As indicated in Table 3, the effect of between level difference of the sample size was not statistically significant (Q-value = .157, p = .69).

3.4.8. Education level

To determine whether the effectiveness of TELL was moderated by education level, the mean effect sizes across different education levels were calculated. The majority of the included studies (N = 40) dealt with undergraduate students, 14 studies considered school students (K-12), and 10 studies examined graduate students. The effect sizes for K-12, undergraduate, and graduate levels were 1.05, .90, and 1.47, respectively. The effect of TELL was large for K-12 students (g = 1.05), undergraduate students (g = .90), and graduate students (g = 1.47). However, the effect of TELL on EFL/ESL writing performance was not moderated by educational level (Q = 3.083, p = .214).

3.5. Meta-regression

To answer the fourth research question examining the relationship between the type of technology and genre of writing (two moderators), a meta-regression was run (Table 4). First, a prediction model was defined and created. Type of technology (I) was coded '0' for collaborative technology and '1' for non-collaborative technology. Conversely, the genre of writing (I) was coded '0' for narrative writing and '1' for argumentative writing. As indicated in Table 4, the coefficient for the type of technology (I) (non-collaborative) was .5039. As Borenstein et al. (2017) explained, coefficients for covariates indicate how covariates are related to the effect size. So, for studies with the narrative genre in non-collaborative technology environments, the effect size was predicted to be .5039 higher than that of environments with collaborative technology. In fact, the predicted effect size of collaborative technology was

Table 4. Main results for the model - Meta-regression results.

Covariate	Coefficient	Standard Error	95% Lower	95% Upper	Z-value	2-sided p-value
Intercept	1.0811	.1690	.7500	1.4123	6.4000	.0000
Non-collaborative (I)	.5039	.1977	−.8914	−.1165	−2.5500	.0108
Argumentative (I)	.1314	.1022	.0545	.9218	2.2100	.02740
Non-collaborative (I) X Argumentative (I)	−.8023	.2814	−.9026	.1953	−3.8100	.0487

1.0811, and the effect size of non-technology environments subsequently equaled 1.584 (according to this formula: 1.0811 + .5039). The difference between them (.5039, p = .01) was equal to the coefficient for the type of technology. So, the results indicated that non-collaborative technologies had a higher effect size for the narrative genre (g = 1.58) than collaborative technologies. Conversely, the coefficient for argumentative writing was .1314, which revealed the effect size for argumentative writing in collaborative environments was .1314 higher than that of the narrative genre. In fact, the predicted effect sizes were 1.081 for narrative and 1.212 for argumentative genres (1.081 + .1314). Therefore, this finding indicated that collaborative technology was more effective for improving argumentative writing (ES = 1.212), while non-collaborative technology was more effective for the narrative genre (ES = 1.584).

To find out whether the relationship between the type of technology and effect size varies by genre of writing, predictor interaction models were considered. As Table 4 demonstrates, the impact of the type of technology on studies with narrative genres was .5039, whereas the impact of type of technology on the argumentative genre was −.2966 (.5039 − .8023). The difference between them was .8023, which was the interaction of type of technology by genre of writing. In addition, the impact of genre of writing on non-collaborative technology was −.6709 (.1314 − .8023), while the impact of the genre of writing on collaborative technology was .1314. The interaction of type of technology by genre of writing was .8023. As indicated in Table 4, the relationship between these two covariates and the treatment effect was stronger than what we would expect by chance, $Q_{model}(3) = 14.16$, $p = .02$ (Table 5). Therefore, the meta-regression model regarding the interaction between type of technology and genre of writing resulted in a significant fit ($p < .05$). As R^2 analog equaled .52, it can be interpreted that 52% of the initial between-study variance in effect sizes can be explained by this combination of covariates.

4. Discussion

The purpose of the present meta-analysis was to examine the overall effectiveness of TELL on EFL/ESL writing performance. In this systematic review, sixty-four studies meeting the inclusion criteria were used for

Table 5. Random-effects model - test of the model.

Test of the model:
Simultaneous test that all coefficients (excluding intercept) are zero
$Q^*_{model} = 14.16$, $df = 3$, $p = .02$
Goodness of fit: Test that unexplained variance is zero
$Tau^2 = .4628$, $Tau = .6883$, $I^2 = 85.02$, $df = 57$, $p = .0000$
R^2 analog = .5214

statistical analysis. Regarding the first research question, the results revealed that TELL had a large positive effect ($g = 1.00$) on EFL/ESL learners' writing performance. Also, the findings indicated that learners who receive technology-integrated instruction perform better on measures of EFL/ESL writing performance than those receiving non-technology writing instruction. The mean effect size obtained in this study ($g = 1.00$) is comparable to the effect size reported in Xu et al. (2019) meta-analysis investigating the effect of collaborative technology on adult learners' writing (ES = .93, 1.28). Based on Cohen's interpretation of the magnitude of effect size, both the effect size of the present study ($g = 1.00$) and the effect size reported by Xu et al. (2019) are large.

One possible explanation for the effectiveness of TELL can be the fact that educational technologies can enhance different aspects of ESL/EFL writing performance such as spelling, structure, and content through various options that they offer; this can lead to greater writing accuracy (Elola & Oskoz, 2017; Hsu & Liu, 2019). Additionally, scholars (Hsu & Liu, 2019; Zhang & Zou, 2021) argue that technology-integrated writing instruction increases learners' engagement with the writing course and their consciousness of the writing process and hence can encourage them to exert greater efforts toward the course. All these might explain why the learners' writing performance improved in a technology-integrated environment.

Considering the second research question, the results of the present study revealed that there is statistically significant heterogeneity among the effect sizes of the meta-analyzed studies. With respect to the third research question, i.e., detecting the potential moderators leading to such variation, eleven sub-group analyses were performed. Among the hypothesized moderator variables, two variables significantly moderated the size of TELL effect, i.e., genre of writing and type of technology. As mentioned earlier, the included studies were classified into seven genres of writing, i.e., academic, argumentative, descriptive, expository, narrative, summary, and mixed genres. The findings revealed that the effect size of argumentative writing ($g = 1.007$) was larger than that of narrative writing. Scholars argue that argumentative writing demands more complex language than narrative essays since argumentative writing requires more reasoning (Shi et al., 2019; Yoon & Polio, 2017). Robinson's cognition hypothesis (Robinson, 2001) states that the more complex a task is, the greater the accuracy will be. This might explain why the learners' accuracy was higher in argumentative writing.

The result also showed the effect size for narrative essays (g = .944) is significantly larger than those of summary writing (g = .19) and expository writing (g = .539). This finding is in line with Chew et al. (2019) and Kim and McCarthy (2020) results, showing that summary writing is the most difficult writing task for many L2 learners. Expository writing involves presenting facts and instructing the readers, which requires learners to have enough background knowledge. Additionally, text construction in expository writing is more difficult, because it requires cognitive and abstract reasoning (Berman & Nir-Sagiv, 2007). These cognitive loads, according to Skehan's (1998) limited attentional capacity model, can cause learners not to pay equal attention to all L2 production aspects to be able to meet the conceptual demands placed on them, and as the results of this meta-analysis show, can lead to less accurate L2 writing performance. This finding is in contrast with Ravid's (2005) and Way et al. (2000) findings, showing that expository essays can result in more complex and accurate language compared to narrative essays. The reason can be attributed to the differences between the nature of the present study (a meta-analysis) and that of Way et al. (2000) and Ravid's (2005) studies, which were experimental studies. However, further experimental studies are needed to come up with more concrete results.

Another significant moderator found in this study was type of technology. The finding revealed that studies with collaborative technologies generated a larger effect size than those with non-collaborative technologies. This finding supported the results of the studies by Bikowski and Vithanage (2016) and Wang (2015), which showed the effectiveness of collaborative online technologies in improving learners' writing quality compared to a non-collaborative environment.

Collaborative technologies can potentially provoke more social and negotiated interactions among language learners (Easley, 2020; Shukor et al., 2014) and hence can create better opportunities for collaborative learning and scaffolding. Therefore, during the wring tasks performed through collaborative technologies, language learners can receive feedback on and assistance with their writing performance, which can help them notice their grammatical and organizational problems (Yang & Meng, 2013) and solve their problems by scaffolding each other (Chen, 2016; Nami & Marandi, 2014). These features, according to the tenets behind the zone of proximal development (ZPD), can push learners to perform beyond their current levels of language proficiency (Richards, 2015) and can cause them to improve their language use, text organization, and reasoning in their writing (Shukor et al., 2014). Another benefit of collaborative technologies is that learners' mistakes can be corrected immediately by their peers; this can help them focus on language forms and grammar (Elola & Oskoz, 2010) and learn how to use L2 structures accurately (Mulligan & Garofalo, 2011). Additionally, collaborative environments are more learner-centered and flexible compared to non-collaborative ones. Therefore, they can increase learners' participation, confidence (De Wever et al., 2015), and autonomy (Kessler & Bikowski, 2010), and can decrease writing apprehension (Cho & Lim, 2017); this can encourage learners to feel safe to share their ideas freely, which can improve learners' writing performance (Perez, 2003; Wu, 2015). These might explain why collaborative technologies can lead to better EFL/ESL writing performance.

Regarding the fourth research question, the finding revealed a statistically significant interaction effect between type of technology and genre of writing. In fact, these two moderators interact to amplify or attenuate each other's effects on the effectiveness of TELL. In collaborative environments, argumentative writing had a higher effect size. In contrast, in non-collaborative environments, narrative writing yielded a higher effect size compared to other genres of writing. The argumentative genre is a cognitively demanding type of writing requiring language learners to use higher-order thinking skills for recognizing and applying persuasive or argumentative text structures (Kim & Pae, 2021); expanding the arguments; and providing support for reasons, which are executed through social interaction and collaboration (Crowell & Kuhn, 2014). As discussed earlier, collaborative environments can provide learners with a better opportunity for collaboration, scaffolding, and critically analyzing each other's writing, which in turn can lead to better argumentative writing (Noroozi et al., 2016). Narrative writing is, conversely, less cognitively demanding, requiring learners to merely recite a fictional, real event, or an experience (Justice et al., 2010). These might explain why there is a significant interaction between collaborative technology environments and argumentative writing. These results can, therefore, support Hafour and Al-Rashidy (2020) study, which revealed that online collaborative environments do not significantly influence students' narrative writing performance when compared to individual online writing.

5. Conclusions and future directions

The current meta-analysis was an attempt to examine the overall effectiveness of TELL on EFL/ESL writing performance. The findings of this study signified that learners receiving technology-integrated writing instruction demonstrated better writing performance than their counterparts who received non-technology writing instruction. The results of moderator analysis also showed that genre of writing and type of technology have a significant interaction with TELL, and hence the effectiveness of TELL can vary based on the genres of writing and the types of technology used. Meta-regression analysis revealed that collaborative technologies had a higher effect size on argumentative writing, while non-collaborative technologies yielded a higher effect size on narrative writing.

The present research can provide several suggestions for future researchers who are interested in technology-integrated writing instruction, and for future meta-analysts who attempt to conduct a synthesis in the field of computer-assisted language learning and writing. For future experimental research, this study revealed that it is essential to investigate the following issues carefully. First, regarding the educational level, more studies are needed to consider K-12 students because the majority of the included studies ($N = 50$) investigated adults (graduate and undergraduate students), and hence there has been little attention to children and teenagers; i.e., digital natives, which can be a moderator variable for the new research. Second, for genre of writing, the effect sizes of summary and expository writings appeared noticeably smaller than those for argumentative, academic, and descriptive genres. Therefore, it is essential to conduct studies to find out how technology-integrated instruction should be used to improve summary and expository genres. Thirdly, it was found that the previous studies did not merely focus on a specific genre of writing. In 64 meta-analyzed studies, a large number of studies ($N = 25$) used mixed writing (a combination of different genres). Therefore, future experimental studies should concentrate on specific writing genres when examining the efficiency of different types of technologies in writing instruction. Moreover, most of the included studies did not provide any information about the cultural backgrounds of language learners, while various cultural backgrounds may play a moderator role. Therefore, it is essential for researchers to focus on language learners' cultural background and mother language. In addition, future researchers are suggested to use more standardized tests or report the reliability and validity of the researcher-made tests, since some primary studies failed to provide any results of validity and reliability of their tests, meaning that they are not objective enough. Furthermore, the present meta-analysis showed that there is an interaction between genre of writing and type of technology, which can influence the effectiveness of technology-enhanced writing instruction. Therefore, more studies are needed to investigate the comparative effects of collaborative versus non-collaborative technologies on a specific genre of writing.

With regards to future meta-analysis, while the findings of this study lend support to the quantitative research on technology-enhanced writing instruction, future research would benefit from synthesizing qualitative studies. As a suggestion for future meta-analysis in this area, meta-analysts are advised to synthesize qualitative studies on ESL/EFL writing performance to elaborate on ESL/EFL learners' attitudes, learning strategies, and motivational currents, and also to broaden the scope of technology-enhanced writing Moreover, there is a dearth of meta-analysis on the effectiveness of specific online tools such as Google Docs, Etherpad, and Edmodo, which are commonly used in writing classes. Another line of research can be a systematic review of the articles on TELL in languages other than English. Efforts were made to include all the studies meeting the inclusion criteria in this study. However, the included studies were written in English, and a systematic review of articles written in languages other than English might also present interesting results.

5.1. Pedagogical implications

The findings of this study can provide important pedagogical implications. First, the results of this systematic review can help policymakers and curriculum developers get insights into how TELL can be used to develop language learners' writing skills, which is one of the most challenging skills for both instructors and learners. The findings of this study also put importance on substantive and methodological factors that can influence language learners' writing performance in technology-mediated environments, and need to be considered by curriculum developers and educational practitioners in designing such courses. Second, this study can reorient scholars' research agenda towards unexplored and underexplored areas of technology-enhanced writing instruction. Third, the results of the present study can inform language instructors about different types of educational technologies (i.e., collaborative technology/non-collaborative technology) and how they can improve language learners' writing performance. Third, this meta-analysis provides evidence that specific genres of writing can be improved through the use of specific kinds of technologies; the conclusions that emerged from this study revealed the importance of focusing on a specific genre of writing in technology-mediated classes. Furthermore, this study recommends language teachers increase the effectiveness of technology-enhanced instruction by taking the required genre of writing into consideration before engaging in the detailed planning of their writing courses. For example, they are advised to use online collaborative tools for teaching argumentative writing. Finally, this study may have pivotal pedagogical implications for designers of educational technology to create more artificially intelligent and multi-functional online tools or applications by integrating some paralanguage activities, i.e., audio and video discussion chat. These paralanguage activities increase collaborative learning and can compensate lack of social context cues in online environments, which subsequently lead to a higher level of social presence and better performance (Phirangee & Malec, 2017).

Disclosure statement

The authors declare no potential conflicts of interests with respect to the authorship or publication of this paper.

Notes on contributors

Masumeh Sadat Seyyedrezaei holds a Ph.D. in applied linguistics and is currently a lecturer at the University of Isfahan, Iran. Her main research interests include Computer-Assisted Language Learning (CALL), language teaching and learning, second language writing, sociolinguistics, and meta-analysis.

Mohammad Amiryousefi is a full-time Associate Professor in Applied Linguistics (TESOL) at the University of Isfahan, Iran. His primary areas of teaching are research design for applied linguistics and language teaching methodology. His main research interests are in the areas of language teaching and learning, second language skills, CALL, and TBLT.

Ana Gimeno-Sanz is a Professor of English and Applied Linguistics at Universitat Politècnica de València, Spain. She was the president of EUROCALL, the European Association for Computer-Assisted Language Learning. She is head of UPV's CAMILLE Research Group, devoted to research in CALL. Her areas of research are language learning and teaching, more specifically in the fields of English for Specific Purposes (ESP), Computer-Assisted Language Learning (CALL), and Content and Language Integrated Learning (CLIL).

Mansoor Tavakoli is a Professor of Applied Linguistics at the University of Isfahan, Iran. He has been teaching TEFL courses at the University of Isfahan at B.A., M.A., and Ph.D. levels for almost two decades. His research interests include: second language acquisition, language teaching and assessment, and EAP education.

ORCID

Masumeh Sadat Seyyedrezaei http://orcid.org/0000-0001-7408-0216
Mohammad Amiryousefi http://orcid.org/0000-0003-3779-1523
Ana Gimeno-Sanz http://orcid.org/0000-0003-3366-0729
Mansoor Tavakoli http://orcid.org/0000-0002-4029-466X

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Appendix A

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Appendix B

Table B1. Coding scheme.

Variables	Description
Research design	
True-experimental	Participants are randomly assigned to either the treatment or the control group
Quasi-experimental	There is no random assignment
Measurement type	
Standardized	Well-established national or international standardized test, i.e., TOEFL, IELTS
Researcher-made	Developed by the researcher for the purpose of the study
Type of technology	
Collaborative	Technologies providing the learners with opportunity to interact or collaborate with their teachers or peers.
Non-collaborative	The educational technologies or technology-enhanced environments in which there is no interaction.
Proficiency level	
Elementary	
Lower-intermediate	
Intermediate	
Upper-intermediate	
Advanced	
Treatment duration	
Short	< 12 weeks (3 months)
long	> = 12 weeks
Sample size	
Small	< 60 participants
Large	> = 60 participants
Education level	
K-12	kindergarten (*K*) for 6-year olds through twelfth grade (12) for 18-year olds.
Undergraduate	University students
Graduate	
Genre of writing	
Academic	This genre requires clear, concise, focused, structured evidence. Its purpose is to aid the reader's understanding.
Argumentative	A piece of writing that takes a stance on an issue. And a writer attempts to persuade readers by stating their reasoning and providing evidence.
Descriptive	An essay which requires to describe something or an experience
Expository	A genre of writing which tends to evaluate the evidence, explain, illustrate, or explicate something in a way that it becomes clear for readers.
Narrative	An essay that aims to tell a story.
Summary	A genre of writing which requires the writer to convey a deeper understanding of the overall argument rather than simply paraphrase specific ideas of the text.
Mixed	Different genres of writing

Appendix C

Table C1. Coding scheme/ basic features of the included studies.

Study	Type of technology	Genre of writing	Treatment duration	Research design	Measurement type	Proficiency level	Context	Sample size	Education level
Abdallah and Mansour	collaborative	mixed	8 weeks	quasi-experimental	researcher	intermediate	EFL	20	undergraduate
Abdelrahman et al.	non-collaborative	mixed	4 weeks	true-experimental	researcher	elementary	EFL	28	K-12
Adas and Bakir	collaborative	mixed	8 weeks	true-experimental	researcher	intermediate	EFL	60	undergraduate
Akbari and Erfani	collaborative	mixed	14 weeks	quasi-experimental	researcher	intermediate	EFL	81	graduate
Akbari and Erfani	non-collaborative	mixed	14 weeks	quasi-experimental	researcher	intermediate	EFL	81	graduate
Al-haq and Al-Sobh	non-collaborative	mixed	8 weeks	quasi-experimental	researcher	lower-intermediate	EFL	122	K-12
Alharbi (2015)	collaborative	argumentative	16 weeks	true-experimental	researcher	advanced	EFL	60	undergraduate
Al-Jarf (2004)	non-collaborative	mixed	14 weeks	quasi-experimental	researcher	advanced	EFL	113	undergraduate
Almasri, et al.	collaborative	argumentative	6 weeks	quasi-experimental	researcher	lower-intermediate	EFL	63	K-12
Alshahrani	collaborative	mixed	14 weeks	quasi-experimental	researcher	intermediate	ESL	46	graduate
Alshumaimeri	collaborative	descriptive	6 weeks	quasi-experimental	researcher	elementary	EFL	42	undergraduate
Arslan	collaborative	mixed	28 weeks	quasi-experimental	researcher	advanced	EFL	59	graduate
Arslan and Şahin-Kızıl	collaborative	mixed	8 weeks	quasi-experimental	researcher	intermediate	EFL	50	undergraduate
Bahrami Shams-Abadi et al	collaborative	mixed	12 weeks	quasi-experimental	standardized	advanced	EFL	40	undergraduate
Bailey and Judd	collaborative	argumentative	8 weeks	quasi-experimental	standardized	intermediate	EFL	65	undergraduate
Bamanger and Alhassan	non-collaborative	mixed	6 weeks	quasi-experimental	researcher	not-mentioned	EFL	55	undergraduate
Chuo (2007)	collaborative	mixed	14 weeks	quasi-experimental	researcher	lower intermediate	EFL	103	undergraduate
Ciftci and Kocoglu	collaborative	mixed	14 weeks	quasi-experimental	researcher	intermediate	EFL	30	undergraduate
Ebadi and Rahimi (2017)	collaborative	descriptive	14 weeks	quasi-experimental	standardized	upper intermediate	EFL	20	graduate
Ebadi and Rahimi (2018)	collaborative	argumentative	14 weeks	quasi-experimental	standardized	upper intermediate	EFL	20	graduate
Ekmekci (2017)	collaborative	argumentative	15 weeks	true-experimental	researcher	intermediate	EFL	43	K-12
Erice, and Ertas	collaborative	mixed	10 weeks	true-experimental	standardized	elementary	EFL	47	undergraduate
Heidari et al.	non-collaborative	narrative	5 weeks	quasi-experimental	researcher	intermediate	EFL	60	undergraduate
Hosseinpour et al.	collaborative	mixed	5 weeks	quasi-experimental	researcher	intermediate	EFL	53	undergraduate
Hsu and Lo	collaborative	expository	9 weeks	quasi-experimental	researcher	intermediate	EFL	52	undergraduate
Hsu and Lo	non-collaborative	expository	9 weeks	quasi-experimental	researcher	intermediate	EFL	52	undergraduate
Hung and Young	non-collaborative	academic	20 weeks	true- experimental	researcher	intermediate	EFL	23	graduate
Hwang, et al.	collaborative	descriptive	12 weeks	quasi-experimental	researcher	elementary	EFL	59	K-12
Karami, et al.	non- collaborative	argumentative	16 weeks	quasi-experimental	standardized	advanced	EFL	157	undergraduate
Kargozari and Ghaemi	non- collaborative	academic	10 weeks	quasi-experimental	researcher	advanced	EFL	52	undergraduate
Kashani, et al.	non- collaborative	argumentative	8 weeks	true-experimental	standardized	not mentioned	EFL	64	undergraduate
Keshta, and Harb	non- collaborative	mixed	8 weeks	quasi-experimental	researcher	elementary	EFL	40	K-12
Kioumarsi, et al.	collaborative	argumentative	8 weeks	true-experimental	researcher	intermediate	EFL	16	graduate
Kovach, et al.	collaborative	argumentative	14weeks	quasi-experimental	researcher	advanced	ESL	28	undergraduate
Lam et al.	collaborative	argumentative	7 weeks	quasi-experimental	researcher	elementary	EFL	72	K-12

(*Continued*)

Table C1. Continued.

Study	Type of technology	Genre of writing	Treatment duration	Research design	Measurement type	Proficiency level	Context	Sample size	Education level
Lam et al.	non-collaborative	argumentative	7 weeks	quasi-experimental	researcher	elementary	EFL	72	K-12
Lee et al.	non- collaborative	argumentative	12 weeks	true-experimental	researcher	advanced	EFL	28	undergraduate
Lin (2014)	non-collaborative	mixed	36 weeks	quasi-experimental	standardized	intermediate	ESL	50	undergraduate
Lin et al.	non- collaborative	argumentative	16 weeks	quasi-experimental	researcher	not-mentioned	EFL	68	undergraduate
Liou (1993)	non-collaborative	descriptive	14 weeks	quasi-experimental	researcher	intermediate	EFL	39	undergraduate
Liou et al.	non-collaborative	descriptive	10 weeks	quasi-experimental	researcher	elementary	EFL	42	undergraduate
Malek Mohammadi and Talebinejad	collaborative	academic	10 weeks	quasi-experimental	researcher	upper- intermediate	EFL	90	undergraduate
Marandi and Seyyedrezaie	collaborative	argumentative	14 weeks	quasi-experimental	standardized	intermediate	EFL	84	undergraduate
Moradi and Karimpour	collaborative	descriptive	14 weeks	quasi-experimental	researcher	intermediate	EFL	60	undergraduate
Nazari and Niknezhad	non-collaborative	mixed	4 weeks	true-experimental	standardized	intermediate	EFL	50	graduate
Park and Cho	collaborative	argumentative	14 weeks	true-experimental	researcher	advanced	EFL	32	undergraduate
Ping and Maniam	collaborative	mixed	3weeks	true-experimental	researcher	lower-intermediate	ESL	30	K-12
Saeedi and Meihami	non-collaborative	argumentative	8 weeks	quasi-experimental	researcher	intermediate	EFL	34	undergraduate
Seifeddin et al.	non-collaborative	narrative	12 weeks	quasi-experimental	researcher	elementary	EFL	62	K-12
Seyyedrezaie et al.	collaborative	mixed	20 weeks	quasi-experimental	standardized	intermediate	EFL	96	undergraduate
Shen (2013)	non-collaborative	mixed	18 weeks	quasi-experimental	researcher	intermediate	EFL	100	undergraduate
Suwantarathip and Wichadee	collaborative	mixed	14 weeks	quasi-experimental	researcher	intermediate	EFL	80	undergraduate
Talakoub (2018)	collaborative	mixed	12 weeks	quasi-experimental	standardized	intermediate	EFL	60	K-12
Tezci and Dikici	non-collaborative	narrative	16 weeks	quasi-experimental	researcher	elementary	EFL	52	K-12
Topacio	non-collaborative	argumentative	10 weeks	quasi-experimental	researcher	lower-intermediate	EFL	45	K-12
Torabi and Safdari	non-collaborative	mixed	20 weeks	true-experimental	standardized	intermediate	EFL	97	graduate
Tsai (2015)	non-collaborative	academic	18weeks	quasi-experimental	researcher	advanced	EFL	247	undergraduate
Vijaya Kumar and Shahin Sultana	non-collaborative	argumentative	5 weeks	true-experimental	standardized	intermediate	ESL	60	undergraduate
Wang (2015)	collaborative	argumentative	12 weeks	quasi-experimental	researcher	intermediate	EFL	48	undergraduate
Wichadee	collaborative	summary	14 weeks	true-experimental	researcher	not-mentioned	EFL	80	undergraduate
Yang et al.	non-collaborative	descriptive	12 weeks	quasi-experimental	standardized	elementary	EFL	32	undergraduate
Zaid	non-collaborative (multi-media)	argumentative	12 weeks	quasi-experimental	standardized	upper-intermediate	EFL	108	undergraduate
Zaid	non-collaborative (online reading)	argumentative	12 weeks	quasi-experimental	standardized	upper-intermediate	EFL	108	undergraduate